Plasmid miniprep (alkaline lysis)

Purpose: isolate high-copy plasmid DNA from 1.5–5 ml of *E. coli* overnight culture. Yields 5–20 µg of DNA, suitable for restriction digests, sequencing and transfection. This text is long enough that it wraps onto a second line and maybe a third one, so that we can check how paragraphs are joined together.

Materials

Reagent	Composition	Storage
Buffer P1	50 mM Tris-HCl pH 8.0, 10 mM EDTA, 100 µg/ml RNase A	4 °C
Buffer P2	200 mM NaOH, 1% SDS	Room temp.
Buffer N3	3 M potassium acetate, pH 5.5	Room temp.

Procedure

Lysis

1. Pellet 1.5 ml of overnight culture at 8,000 × g for 2 min at room temperature. Discard the supernatant.
2. Resuspend the pellet in 250 µl Buffer P1 by vortexing until no clumps remain.
3. Add 250 µl Buffer P2 and invert gently 4–6 times. Do not vortex. Incubate no longer than 5 min.

Neutralisation and binding

4. Add 350 µl Buffer N3 and invert immediately 4–6 times.
5. Centrifuge at 17,900 × g (13,000 rpm) for 10 min in a table-top microcentrifuge.

Notes

- Warm Buffer P2 to 37 °C if a precipitate has formed.
- Use 10^6 cells per prep; CO_2 is not required.
- Store the eluted DNA at −20 °C.

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Safety

Buffer P2 contains sodium hydroxide, which is corrosive. Wear gloves and eye pro-tection when handling it. Dispose of the waste accord-ing to local rules.